

CONTACT-FREE BLOOD PRESSURE WAVEFORM ESTIMATION FROM RADAR SIGNALS VIA MULTIMODAL DICTIONARY LEARNING

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ABSTRACT

In this paper, we investigate contact-free blood pressure (BP) monitoring using a chest radar. We propose a dictionary-learning framework that leverages multimodal recordings including radar, ICG, ECG, and BP to learn shared physiological representations. After training, BP waveforms are reconstructed by weighting the learned BP dictionary with sparse codes estimated solely from the radar signal. The learned representation enables interpretable cardiac coding from radar signals alone, allowing the synthesis of high-fidelity BP waveforms. Experiments on public datasets demonstrate interpretable modeling and accurate BP inference, enabling practical single-sensor BP monitoring with multimodal supervision.

Index Terms— Continuous Non-invasive Blood Pressure Monitoring, FMCW radar, Multimodal Dictionary Learning, Sparse Coding

1. INTRODUCTION

Continuous, noninvasive monitoring of arterial blood pressure (BP) waveforms is critical for early cardiovascular risk detection and overall health assessment [1, 2]. A common continuous, noninvasive method is Continuous Noninvasive Arterial Pressure (CNAP): a finger-cuff volume-clamp that yields beat-to-beat waveforms with periodic brachial-cuff calibration [3]. However, the contact cuff and sustained pressure hinder long-term wear and may cause discomfort. This motivates contactless sensing. Frequency-modulated continuous-wave (FMCW) radar, which can detect tiny chest-wall movements caused by cardiac activity, has emerged as a promising candidate for unobtrusive physiological sensing for BP [4, 5].

Despite its potential, estimating the full BP waveform from radar remains challenging. Cardiac-induced motions are extremely subtle and are easily obscured by respiration and other noise sources. Moreover, radar measures chest or neck

displacement, not pressure, so the mapping to the underlying BP waveform is indirect, subject-specific, and highly nonlinear. Consequently, many studies sidestep waveform recovery and only regress scalar indices (SBP/DBP) from pulse-timing features such as pulse arrival time (PAT)[6] and pulse transit time (PTT/R-PTT)[7, 8, 9]. Others employ black-box deep learning, such as end-to-end waveform synthesis [10, 11], but such models often require large datasets and offer limited physiological interpretability, hindering clinical trust.

Recovering the full arterial waveform is critical, as it preserves richer hemodynamic information beyond scalar indices and provides physiological interpretability that is essential for clinical adoption [12, 13]. In this regard, explainable approaches that estimate waveforms rather than black-box regression of BP values may foster greater trust and diagnostic utility in practice [14, 15].

We achieve BP waveform reconstruction from a radar signal in a convolutional sparse coding (CSC) framework inspired by multimodal learning [16, 17, 18, 19]. Specifically, we leverage *synchronized* FMCW-deduced chest displacement, BP waveform, electrocardiogram (ECG), and impedance cardiogram (ICG)¹ to train modality-specific dictionaries that encode cardiac-cycle morphology, along with shared beat-synchronous codes. In inference, only radar signals are required: the BP waveform is estimated via the learned BP dictionary and the sparse codes solely derived from the radar signal.

Compared to prior radar-based BP estimation, our contributions are threefold:

- **First multimodal CSC framework for radar-based BP waveform recovery:** We introduce a multimodal CSC framework that learns a unified physiological representation from synchronized radar, ECG, ICG, and BP during training, while requiring only radar at deployment for accurate BP waveform synthesis.
- **First dictionary-based, interpretable model:** Our model explicitly separates cardiac influences from the noisy signal. The learned dictionary atoms correspond to meaningful features of the cardiac cycle (e.g. systolic peaks, diastolic notch), enhancing transparency.

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¹These signals can be measured by an FMCW radar, a clinical ECG/ICG module, and CNAP; a sample dataset is available at [20].

- **Efficient optimization scheme:** We implement an alternating optimization algorithm in the frequency domain (combining ADMM with convolutional dictionary updates) for tractable training of the CSC model and fast inference on new radar signals.

2. PROPOSED METHOD

This section first introduces the multimodal signal model based on shared sparse codes and modality-specific dictionaries. Next, a multimodal learning strategy is elaborated to obtain these modality-specific dictionaries. Finally, new BP waveforms are reconstructed using the learned BP dictionary and the sparse codes extracted solely from the radar signal with the learned radar dictionary.

2.1. Multimodal Signal Model

Our proposed model is based on the assumption that a common, beat-synchronous cardiac drive gives rise to multiple observations that are expressed through modality-specific transfer functions and delays. In particular, its mechanical activity produces chest-wall motion sensed by radar and pulsatile changes recorded as the BP waveform. Its electrical activity gives rise to signals captured by ECG and to thoracic impedance changes measured by ICG. This latent cardiac drive is modeled as sparse codes, regarded as the response to each cardiac cycle, while the corresponding modality-specific transfer functions and delays are modeled as convolutional dictionaries. This assumption is supported by the cross-modal alignment observed in Fig. 2(a) of [20].

We denote the radar signal as $\mathbf{s}_{rad} \in \mathbb{R}^N$, ECG as $\mathbf{s}_{ecg} \in \mathbb{R}^N$, ICG as $\mathbf{s}_{icg} \in \mathbb{R}^N$, and reference BP as $\mathbf{s}_{bp} \in \mathbb{R}^N$, all sampled at N time points. The multimodal signal model is given by

$$\begin{aligned}
 \text{(Radar)} \quad \mathbf{s}_{rad} &= \sum_{k=1}^K \mathbf{d}_{rad,k} * \mathbf{c}_k + \boldsymbol{\epsilon}_{rad}, \\
 \text{(ECG)} \quad \mathbf{s}_{ecg} &= \sum_{k=1}^K \mathbf{d}_{ecg,k} * \mathbf{c}_k + \boldsymbol{\epsilon}_{ecg}, \\
 \text{(ICG)} \quad \mathbf{s}_{icg} &= \sum_{k=1}^K \mathbf{d}_{icg,k} * \mathbf{c}_k + \boldsymbol{\epsilon}_{icg}, \\
 \text{(BP)} \quad \mathbf{s}_{bp} &= \sum_{k=1}^K \mathbf{d}_{bp,k} * \mathbf{c}_k + \boldsymbol{\epsilon}_{bp},
 \end{aligned}$$

where $*$ denotes convolution. Each $\mathbf{d}_{(m,k)} \in \mathbb{R}^L$ is the k -th dictionary atom for the cardiac signal, $m \in \mathcal{M} \doteq \{\text{rad}, \text{ecg}, \text{icg}, \text{bp}\}$, and $\mathbf{c}_k \in \mathbb{R}^N$ is the k -th *shared* sparse code capturing the beat-synchronous cardiac activity. We use K atoms per modality to capture distinct morphological components of the cardiac cycle (e.g., onset, systolic peak,

dicrotic notch). The noise term $\boldsymbol{\epsilon}_{(m)} \in \mathbb{R}^N$ account for measurement and modeling errors.

2.2. Training: Dictionary Learning

For notational convenience, we further denote

$$\mathbf{D}_m \doteq [\mathbf{d}_{m,1}, \mathbf{d}_{m,2}, \dots, \mathbf{d}_{m,K}]^T \in \mathbb{R}^{K \times L}, \quad (1)$$

$$\mathbf{c} \doteq [\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_K]^T \in \mathbb{R}^{K \times N}, \quad (2)$$

where $m \in \mathcal{M} \doteq \{\text{rad}, \text{ecg}, \text{icg}, \text{bp}\}$. For the long-term data, we divide it into multiple windows. Consider the j -th window $\mathbf{s}_m^{(j)}$ of the data, the dictionaries $\mathbf{D}_m$ s are shared across all windows, capturing how that activity manifests in each sensor's signal, while the sparse codes are window-specific, reflecting the common cardiac drive, denoted as $\mathbf{c}^{(j)}$. Each signal model in window j can be compactly written as

$$\mathbf{s}_m^{(j)} = \mathbf{D}_m * \mathbf{c}^{(j)} + \boldsymbol{\epsilon}_m^{(j)}. \quad (3)$$

The overall optimization problem in the training stage is

$$\begin{aligned}
 \min_{\{\mathbf{D}_m, \mathbf{c}^{(j)}\}} \quad & \sum_{j=1}^J \sum_m \|\mathbf{s}_m^{(j)} - \mathbf{D}_m * \mathbf{c}^{(j)}\|_2^2 + \lambda \mathcal{R}(\mathbf{c}^{(j)}), \\
 \text{s.t.} \quad & \text{diag}(\mathbf{D}_m^\top \mathbf{D}_m) = \mathbf{1}_K, \quad \forall m,
 \end{aligned} \quad (4)$$

where $m \in \mathcal{M}$, $\mathbf{s}_m$ is the signal for modality m , and $\mathbf{D}_m$ is the corresponding dictionary. The constraint on $\mathbf{D}_m$ ensures each atom $\mathbf{d}_{m,k}$ has unit norm to avoid scale ambiguity. To reduce solution multiplicity, we upgrade the sparsity prior on the codes to a group-sparsity penalty, $\mathcal{R}(\mathbf{c}^{(j)})$, which can be expressed as

$$\mathcal{R}(\mathbf{c}^{(j)}) \doteq \|\mathbf{c}^{(j)}\|_{2,1} = \sum_{n=1}^N \|\mathbf{c}_{(n)}^{(j)}\|_2, \quad (5)$$

where $\mathbf{c}_{(n)}^{(j)}$ is the n -th column of $\mathbf{c}^{(j)}$. This penalty promotes that the non-zero elements of $\mathbf{c}_k$ s emerge at the same time points across different atoms indexed by k , naturally leading to the sparsity for each $\mathbf{c}_k$, which aligns with the physiological prior that cardiac events are sparse in time.

We solve this non-convex problem using an alternating minimization scheme, iterating between two main steps.

2.2.1. Sparse Code Update

With all the dictionaries fixed in (4), we update the group-sparse codes $\mathbf{c}$ for each windowed segment of data. The optimization problem for a single window is:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \sum_m \frac{1}{2} \|\mathbf{s}_m - \mathbf{D}_m * \mathbf{c}\|_2^2 + \lambda \sum_{n=1}^N \|\mathbf{z}\|_{2,1} \quad \text{s.t.} \quad \mathbf{c} = \mathbf{z}. \quad (6)$$

Algorithm 1 ADMM for CSC problem (6)

Input: For a single window: modality signals $\{\mathbf{s}_m^{(j)}\}_{m \in \mathcal{M}}$; fixed dictionaries $\{\mathbf{D}_m\}_{m \in \mathcal{M}}$ with K atoms; penalty $\rho > 0$; sparsity weight $\lambda > 0$; iterations T .

Output: Sparse codes $\mathbf{c}^{(j)} \in \mathbb{R}^{K \times N}$.

1: **Preprocessing**

For each $m \in \mathcal{M}$, let $N_f = \lfloor N/2 \rfloor + 1$ and compute

$$\hat{\mathbf{D}}_m \leftarrow \text{RFFT}(\text{pad}_N(\mathbf{D}_m)) \in \mathbb{C}^{K \times N_f},$$

$$\hat{\mathbf{s}}_m \leftarrow \text{RFFT}(\mathbf{s}_m^{(j)}) \in \mathbb{C}^{N_f}.$$

Initialize $\mathbf{C}_\omega, \mathbf{Z}_\omega, \mathbf{Y}_\omega \in \mathbb{C}^{K \times N_f}$ to zeros, where ω indexes frequency bins.

Precompute $\mathbf{G}_f = \sum_{m \in \mathcal{M}} \hat{\mathbf{D}}_{m,f} \hat{\mathbf{D}}_{m,f}^H \in \mathbb{C}^{K \times K}$ for $f = 0, \dots, N_f - 1$, where $\hat{\mathbf{D}}_{m,f} \triangleq \hat{\mathbf{D}}_m(:, f)$.

2: **for** $t = 1$ to T **do**

3: **Per-frequency linear solve:**

$$\mathbf{r}_f \leftarrow \sum_{m \in \mathcal{M}} \hat{\mathbf{D}}_{m,f} \hat{\mathbf{s}}_m[f] + \rho(\mathbf{Z}_\omega[:, f] - \mathbf{Y}_\omega[:, f]),$$

$$\mathbf{C}_\omega[:, f] \leftarrow (\mathbf{G}_f + \rho \mathbf{I})^{-1} \mathbf{r}_f.$$

4: **Proximal Update for ℓ_1 in time:**

$$\mathbf{C}_{\text{time}} \leftarrow \text{IRFFT}(\mathbf{C}_\omega + \mathbf{Y}_\omega) \in \mathbb{R}^{K \times N},$$

$$\mathbf{Z}_{\text{time}}[:, n] \leftarrow \left(1 - \frac{\lambda/\rho}{\|\mathbf{C}_{\text{time}}[:, n]\|_2 + \varepsilon}\right)_+ \mathbf{C}_{\text{time}}[:, n], \forall n,$$

$$\mathbf{Z}_\omega \leftarrow \text{RFFT}(\mathbf{Z}_{\text{time}}).$$

5: **Dual update:**

$$\mathbf{Y}_\omega \leftarrow \mathbf{Y}_\omega + (\mathbf{C}_\omega - \mathbf{Z}_\omega).$$

6: **return** $\mathbf{c}^{(j)} = \mathbf{z}_{\text{time}}$.

This is a standard CSC problem, which we solve efficiently in the frequency domain using the Alternating Direction Method of Multipliers (ADMM) [21]. In practice, we use the Real Fast Fourier Transform (RFFT) to map the real-valued time-domain signals and dictionaries into the frequency domain, and the Inverse RFFT (IRFFT) to recover the updated coefficients back into the time domain. See Algorithm 1 for details.

2.2.2. Dictionary Update

With the sparse code $\mathbf{c}$ fixed, we update the dictionaries $\{\mathbf{D}_m\}$. This is a series of least-squares problems, one for each dictionary. Again, we solve this efficiently in the frequency domain using a vectorized approach known as Convolutional Coordinate MOD (CCMOD) [22]. After each update, the dictionary atoms are projected back to the unit ball to prevent scale ambiguity; see Algorithm 2 for details.

Algorithm 2 Batched CCMOD Dictionary Update

Input: For window j : modality signals $\{\mathbf{s}_m^{(j)}\}_{m \in \mathcal{M}}$; fixed codes $\mathbf{c}^{(j)} \in \mathbb{R}^{K \times N}$; ridge weight $\mu \geq 0$.

Output: Updated dictionaries $\{\mathbf{D}_m\}_{m \in \mathcal{M}}$.

1: **Preprocessing**

$$\hat{\mathbf{c}}_\omega \leftarrow \text{RFFT}(\mathbf{c}^{(j)}) \in \mathbb{C}^{K \times N_f},$$

$$\hat{\mathbf{s}}_m \leftarrow \text{RFFT}(\mathbf{s}_m^{(j)}) \in \mathbb{C}^{1 \times N_f},$$

$$\hat{\mathbf{s}}_{cc}[f] \leftarrow \sum_b \hat{\mathbf{c}}_\omega[b, :, f] \hat{\mathbf{c}}_\omega[b, :, f]^H \in \mathbb{C}^{K \times K},$$

$$\hat{\mathbf{s}}_{sc,m}[f] \leftarrow \sum_b \hat{\mathbf{s}}_m[b, f] \hat{\mathbf{c}}_\omega[b, :, f]^H \in \mathbb{C}^{1 \times K}.$$

2: **for each** modality $m \in \mathcal{M}$ **do**

3: **for** For each frequency bin $f \in \{0, \dots, N_f - 1\}$ **do**

$$4: \quad \hat{\mathbf{D}}_m[:, f] \leftarrow (\hat{\mathbf{s}}_{cc}[f] + \mu \mathbf{I})^{-1} \hat{\mathbf{s}}_{sc,m}[f]^H.$$

$$5: \quad \tilde{\mathbf{D}}_m \leftarrow \text{IRFFT}(\hat{\mathbf{D}}_m) \in \mathbb{R}^{K \times N}$$

$$6: \quad \mathbf{D}_m \leftarrow \tilde{\mathbf{D}}_m[:, 0:L]$$

$$7: \quad \mathbf{d}_{m,k} \leftarrow \mathbf{d}_{m,k} / \|\mathbf{d}_{m,k}\|_2$$

8: **return** $\{\mathbf{D}_m\}_{m \in \mathcal{M}}$.

2.2.3. Training Strategy

We adopt a three-stage curriculum to cope with the radar signal of low quality and the relatively clean ECG/ICG/BP channels. Each stage contains multiple outer iterations, in each of which codes $\mathbf{c}$ are first updated via Algorithm 1, and then dictionaries are updated via Algorithm 2. In each stage, different codes and dictionaries are updated.

- **Stage A (bootstrap on clean modalities; T_a outer iters.):** Use only ECG and ICG to update both the shared code $\mathbf{c}$ and their dictionaries. This stage yields a sparse, quasi-periodic code without explicitly enforcing periodicity.
- **Stage B (transfer to difficult modalities; T_b outer iters.):** Introduce radar and BP but *keep $\mathbf{c}$ updated only from ECG/ICG*. Thus $\mathbf{c}$ remains stable and high-SNR, and $\mathbf{D}_{\text{rad}}, \mathbf{D}_{\text{bp}}$ are fitted to that code via CCMOD.
- **Stage C (joint fine-tuning; T_c iters.):** Unfreeze all modalities and jointly update codes and dictionaries. This stage co-adapts radar/BP with the stabilized code, aligns cross-modal scales/timing, and removes residual mismatch from Stages A-B.

2.3. Inference: Radar-Only BP Estimation

Once the dictionaries $\mathbf{D}_{\text{rad}}$ and $\mathbf{D}_{\text{bp}}$ are learned, the other modalities are no longer needed. For a new, unseen radar signal $\mathbf{s}_{\text{rad}}$, we perform the following two steps for inference:

1. **Estimate Cardiac Code:** We solve the following problem via Algorithm 1 to find the cardiac code $\hat{\mathbf{c}}$:

$$\hat{\mathbf{c}} = \arg \min_{\mathbf{c}} \frac{1}{2} \|\mathbf{s}_{\text{rad}} - \mathbf{D}_{\text{rad}} * \mathbf{c}\|_2^2 + \lambda_{\text{test}} \mathcal{R}(\mathbf{c}), \quad (7)$$

where $\mathcal{R}(c)$ is the group-sparsity penalty defined in (5).

2. **Synthesize BP Waveform:** We then use the estimated cardiac code $\hat{c}$ and the learned BP dictionary $\mathbf{D}_{bp}$ to synthesize the final BP waveform:

$$\hat{s}_{bp} = \mathbf{D}_{bp} * \hat{c}. \quad (8)$$

3. EXPERIMENTS AND RESULTS

3.1. Dataset and Preprocessing

We evaluate our proposed framework on a public dataset [20]. The dataset consists of simultaneous recordings of radar signals, ECG, ICG, and continuous BP waveforms from 30 subjects under various scenarios. We use the resting state data for our experiments, which includes approximately 10 minutes of data per subject. We construct the model for each single subject, and removed the first 50 seconds to avoid transient effects. Then the 500 seconds of the remaining data were used for training, and the last 50 seconds for testing. All signals were resampled to 50 Hz, and segmented into 10-second windows with a 3-second overlap. The radar signal was preprocessed via the methods the dataset provides to remove motion artifacts and obtain a clean radar pulse signal.

3.2. Results

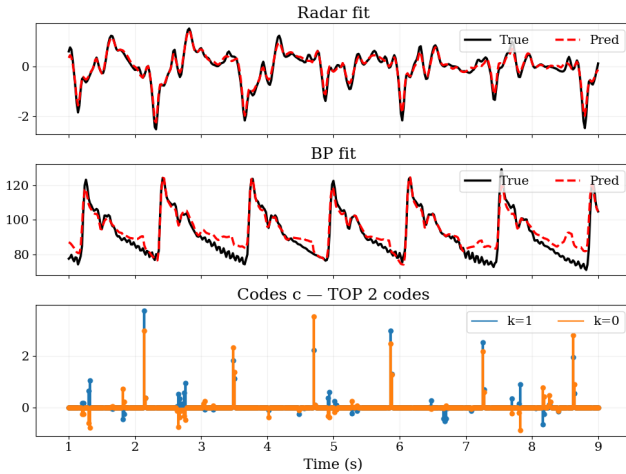


Fig. 1. Fitted waveforms of one window.

Fig. 1 shows an example of the fitted BP waveform from our proposed method compared to the ground truth. The first subplot shows the fitted radar signal, and the second subplot shows the fitted BP waveform. The last subplot shows the codes learned from the radar signal, which captures the cardiac cycles. We can clearly see that the codes are nearly group sparse, with non-zero activations occurring around the systolic peaks of the BP waveform. This indicates that the

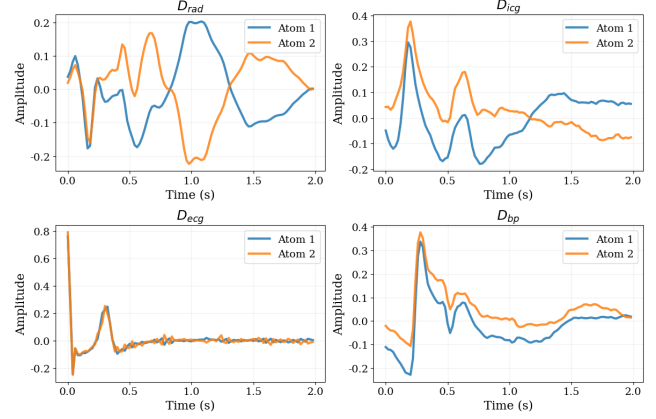


Fig. 2. Visualization of the learned cardiac dictionaries of Radar, ICG, ECG, and BP, demonstrating the interpretability of our model.

learned codes effectively capture the underlying cardiac activity that drives both the radar and BP signals.

Furthermore, the learned dictionaries in our model offer interpretability. Fig. 2 shows the learned cardiac dictionary atoms, demonstrating the model's ability to isolate physiologically meaningful patterns.

Table 1. Quantitative comparison of BP estimation.

Method	SBP (MAE)	DBP (MAE)	BP waveform (MAE)	R
R-PTT regression [9]	8.95	7.53	—	-0.05
Ours (CSC)	6.01	3.38	6.08	0.53

Table 1 summarizes the quantitative results of our proposed method compared to the classic reflective PTT (R-PTT) based regression method. The metrics are Mean Absolute Error (MAE) and Pearson Correlation Coefficient (R). The R-PTT-based method uses the time difference between the two peaks of the radar signal to estimate BP via a linear regression model, which is a common approach in cuffless BP estimation. However, this method highly relies on accurate peak detection, which can be challenging in noisy radar signals. Our proposed CSC method significantly outperforms the R-PTT-based method in terms of SBP and DBP, and enjoys better correlative coefficient **R**. Moreover, our method provides an accurate reconstruction of the entire BP waveform, which is not achievable with the R-PTT-based approach.

4. CONCLUSION

In this paper, we presented a multimodal convolutional sparse coding framework for continuous BP waveform estimation from contact-free radar signals. By leveraging multimodal data during training, our model learns an interpretable, shared physiological representation that allows it to operate on radar signals alone during inference.

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